The Intersection of VR and Education: Exploring the Creation of a VR Educational Escape Room

CSC-475: Seminar in Computer Science
Spring 2024

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Abstract

This project focuses on the intersection between VR and education. Described as an immersive tool, VR helps students become engulfed in environments otherwise unknown or unreachable to them. Through this innovative technology, students are able to apply their learning in new and different ways. This intersection also involves the gamification of educational topics in a multitude of ways. One of these ways is escape rooms. By combining the use of VR technology and escape rooms, I have set out to create a fully immersive form of supplemental material for students. However, during this process, I ran into a few bugs. These included hardware issues as well as a very steep initial learning curve. In the future, the goal is to fully test and implement this "educational escape room" with a group of elementary to middle school age students.

1 Introduction

This project explores VR technology in conjunction with education and the concept of gamification [1] to create an educational escape room. The goals of this project are to do the following:

1. Conduct a literature review on the intersection of VR and education
2. Begin the process of creating an educational escape room in VR using Unity
3. Test the escape room on elementary to middle school age students
4. Evaluate results and compare with conclusions drawn from the literature review

With over 1,900 escape rooms in the US in 2022 and even more globally, escape rooms have become a trend and a popular group activity [2]. With the rise in popularity of VR, I have decided to create a virtual escape room. With this escape room I hope to incorporate various puzzles, games, and an immersive landscape.

Alongside the rise in popularity of escape rooms, came the concept of integrating escape rooms into education. These escape rooms would be used as an immersive experience to let students use their knowledge in an applied manner.[3]

On the topic of immersive experiences in education, comes the discussion on Virtual Reality and its uses in education. While there is a lack of information on the neurological benefits of VR in education, we have seen a steady rise in literature on the various uses of VR in education and the results it's produced. [4, 5, 6] As part of my project I have conducted a literature review on the intersection of VR and education and any implications the literature may claim in the integration of VR in an educational context.

Figure 1: Benefits of VR in Education

Based off the steady increase in the usage of VR and escape rooms in the classroom, I have decided to combine the two and create an educational escape room. This escape room would be catered towards kids around a late elementary/middle school age range and would primarily be used as supplemental material, rather than trying to teach concepts that have not been covered. The goal is to have a fully functional and tested escape room with interactive puzzles an games to help the user escape the room. Each puzzle would be testing a different part of the material being taught.

In this paper I will discuss the literature review I conducted and the major takeaways, the methods of creating the escape room (including any setbacks, problems, or technical issues), preliminary results, and the future implications and steps for this project.

2 Literature Review

Using the search terms "[Virtual Reality OR VR] AND [Education]", "[VR] AND [Escape Rooms]", and "[Escape Rooms] AND [Education]", I was able to not only conduct a literature review on the intersection of VR and education, but also the implementation of escape rooms in education (both traditional and VR). The main focus of this section, however, will be the aforementioned intersection and implementation of Virtual Reality in the classroom.

Before discussing the use of VR in education, it's important to understand the neurological effects that VR is said to have on people. In a paper by Gumilar, et al., a neurological technique known as hyperscanning, was used to "record neural activity" while using VR in a collaborative context. [7] The study ultimately concluded that VR can elicit and "enhance inter-brain synchrony" in a collaborative VE (Virtual Environment) and that they were able to demonstrate that VR could help neurologists develop a more thorough understanding of human communication. [7]

The overall purpose of this literature review is to highlight the researched benefits of VR in education. Thus, we should discuss what those benefits are. We can split up the benefits into 3 separate categories:

1. Enhanced engagement and motivation
2. Immersive and experiential learning opportunities
3. Access to otherwise inaccessible environments or experiences.

For each of these benefits I will explain more and provide context for what sets VR apart from other traditional learning methods.

2.1 Enhanced Engagement and Motivation

Virtual reality is defined by many people as an interactive and immersive experience which engulfs and engages the user in an environment different from their own. [8, 9, 10, 11, 12, 13, 5, 14] Because of this immersion and the added impact of having a headset/headgear on, acting as "blinders", the user is fully engaged by their environment, blocking out the rest of the world around them. The immersive experience of using Virtual Reality has shown to motivate students and capture their attention in ways that may help them retain knowledge and focus better on the material at hand. [1, 15, 16, 11, 5] Compared to the more traditional learning methods, VR provides a new avenue for students to engage with their material, and they are more likely to *stay* engaged with it for the duration of the session.

2.2 Immersive and Experiential Learning Opportunities

As mentioned before, VR is very immersive, and can cause the user to be completely enveloped by their virtual environment [7, 17, 8, 3, 18, 9, 19, 20, 15, 11, 12, 13, 5]. This allows them to "become part" of the material like they are studying, as well as letting them see visualizations of what they are learning in new ways. For example, in a study done, students were able to use a system in which they became gorillas in a habitat and explored the way they live. [21] In a couple of other studies, students took part in virtual laboratories where they were able to engage with molecules and take part in scientific experiments, simply by using VR technology. [12, 22] VR allows students with hands-on experiential learning opportunities that help them better understand and retain material they are being taught. [1]

2.3 Access to Inaccessible Environments or Experiences

Immersion seems to be the highlight of most of these benefits, and there's no difference with this one. Because virtual reality lets the user explore different environments within their own space, it opens up the opportunity for new experiences they may not have had access to before. For example, historical sites, zoo exhibits, landmarks, outer space, etc. VR allows students to transcend the boundaries of the traditional classroom and offers new possibilities and experiences that are otherwise inaccessible to them.

Moving on from the benefits, I want to discuss the concept of escape rooms, both virtual and non-virtual, in education as discussed in the literature. In one study, the researchers created a stereochemistry escape room that combined a traditional escape room with VR technology. [20] In another study, students were given the task of creating an escape room with VR/AR technology in an attempt to use their astronomical knowledge to create a space based escape room. [3] As we can see, although the literature I found is limited, the concept of Virtual Escape rooms is definitely a concept being explored in the education world. One study discussed the results of a survey done on whether or not students found a previously distributed escape room to be beneficial, and the results were highly positive. There has also been limited research done on the use of non-virtual escape rooms in education. [1, 15, 4] These more traditional escape rooms garnered similarly positive reactions from those using them. [1, 15]

Now that we have discussed the benefits of VR in education and the use of escape rooms, and the different ways they are being implemented, we should discuss some drawbacks and limitations discussed in the literature, about the implementation of VR in education.

Some of the main problems include:

1. Lack of Neurological Evidence on Benefits of VR in the Classroom
2. Time to Develop and Learning Curve
3. Accessibility: Special Education
4. Accessibility: Funding

2.4 Lack of Neurological Evidence on Benefits for VR in the Classroom

While VR does seem to be very beneficial in an educational context, that doesn't mean it's all good. In fact, there actually seems to be quite a few problems discussed in the literature. For one, there seems to be an overall lack of literature on the neurological benefits of VR in an educational context. While there have been studies done that report perceived benefits, I think until there is more medically proven results, the rollout of VR in classrooms may remain limited or minimal [7, 23].

2.5 Time to Develop and Learning Curve

Another major drawback is the time it takes to create these educational applications. Much like other applications, it takes a long time to create VR apps, and having to put in educational aspects adds more time to the process. The main reason why, is discussed in a few papers: you need to know the material well enough to actually be able to create something for it. [3, 1, 21] The problem with this leads to the discussion of another drawback, the knowledge gap. [19]

Many educators, because they know little to nothing about VR technology aren't fully convinced about integrating VR into their classrooms. [19] This hesitation is understandable to a certain extent because the learning curve for VR and the development of VR applications is heavy. If a teacher doesn't know what's going on with VR, how would they feel comfortable enough to implement it? And if there's a lack of people who do fall into the small intersection of VR and education, how are the applications supposed to be developed? This learning curve/knowledge gap is a big drawback according to many of the authors featured in the literature review. On this same topic, it's important to note that because of the knowledge gap, many researchers have concluded that VR should only be used as *supplemental* material.

2.6 Accessibility: Special Needs

We also have to discuss issues with accessibility. Some studies have tested out Virtual Reality on students with special needs. [6, 10, 12, 24] While there are success stories and a good amount of the students were able to interact and properly engage with the VE, there is still a worry of, did they actually see anything? Is it overstimulating? [12] And in those same studies, there were students who refused to even put on the headset. [12] So, there is a concern about what would happen with students who are uncomfortable with VR if it became an integral part of education in the future?

2.7 Accessibility: Funding

There is one more drawback I'd like to discuss, and perhaps the most obvious one: funding. VR is no doubt, very expensive. This creates a problem for Title 1 schools and other institutions that may not have the funds to implement VR in their classrooms. [11] While that doesn't seem like a huge set back at this moment, if VR actually does take off, the children and students at these institutions will be at a disadvantage compared to their peers at other schools.

Overall, this literature review provided a great basis and insight into Virtual Reality and education, as well as with the implementation of escape rooms in a learning environment. In the following sections, I will discuss the process of creating the educational escape room, my preliminary results, and future work.

3 Methods

I'll divide my development process into a few steps:

1. Choosing Between Unity and Unreal Game Engines
2. Beginner Tutorials and Setting Up Unity
3. VR Basics Tutorials
4. Creating Puzzles
5. Escape Room Tutorials and Building My Own
6. Running into Problems and Initial Tests
7. Transitioning to a Literature Review and Final Thoughts

For each of these steps I'll go into detail explaining the process and how I went about carrying them out.

3.1 Unity vs. Unreal Game Engines

When it comes to developing VR games and applications, there's two main game engines used: Unity and Unreal. So to start off my process, I had to decide which one to go with. After doing some research on which was better, I ultimately landed on Unity. I chose Unity for a few different reasons, the main selling point being the beginner friendly reputation. Unity is known to be very easy to use and intuitive for beginners, as many of the starter project don't involve any mandatory coding or prior programming experience. And with a multitude of assets and projects with Unity Learn, the technical work is minimal if any. However, Unity sacrifices the quality and performance of it's applications for the smooth User Friendly interface and design.

Figure 2: Disadvantages

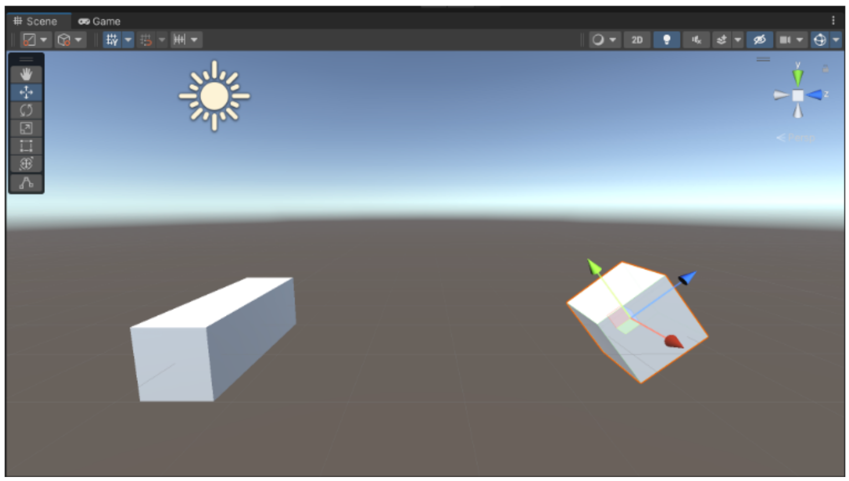

Figure 3: 3D Basics in Unity

For Unreal, it's the opposite. Made for high fidelity graphics and games, Unreal is far from beginner friendly and involves more technical experience than Unity. Even though both provide assets and copious libraries, Unreal is known to have a steeper learning curve than building with Unity. I was willing to risk the performance of my escape room for a more user friendly experience, especially because I had a limited amount of time to get a project up and running.

3.2 Beginner Tutorials and Setting Up Unity

After choosing the game development engine, I moved on to finding tutorials on how to use Unity. Using Unity's Unity Learn platform, I started basic tutorials learning the ins and outs of how to create basic 3d images and figures in Unity [25] 3. The beginning tutorials went smoothly and I was able to follow along pretty well. The basic ideas were easy to follow and understand, and were far from technical or confusing.

3.3 VR Basics Tutorials and Setting Up the Headset

Shortly after finishing Unity Learn's basics pathway, I moved onto the VR basics tutorials. 4 These, once again, were easy and intuitive; even the challenges they had included were simple and are probably achievable for anyone with at least some programming experience. 5

The problems began, however, when the set up process for the headset began. For this project I used a MetaQuest 2 headset. While turning it on and familiarizing myself with the controls was easy, the actual set up process is when things started to go downhill. After trying multiple times to try and get builds to run using developer mode on the headset, I ended up switching over to the Oculus app. Through a USB-3.0 cable and the Oculus app on my PC, I was able to get the headset up and running and test out the VEs made from the tutorials.

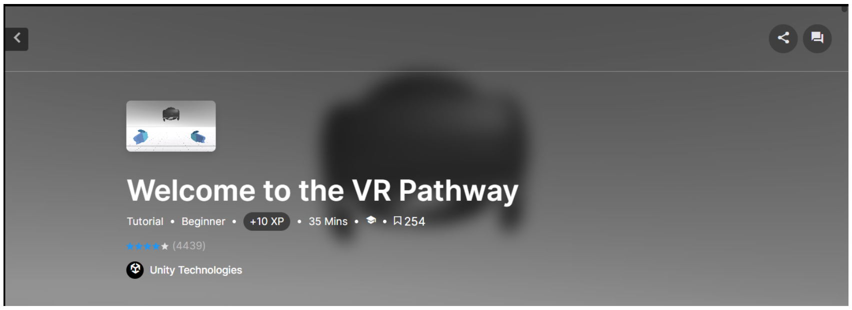

Figure 4: VR Tutorial on Unity Learn

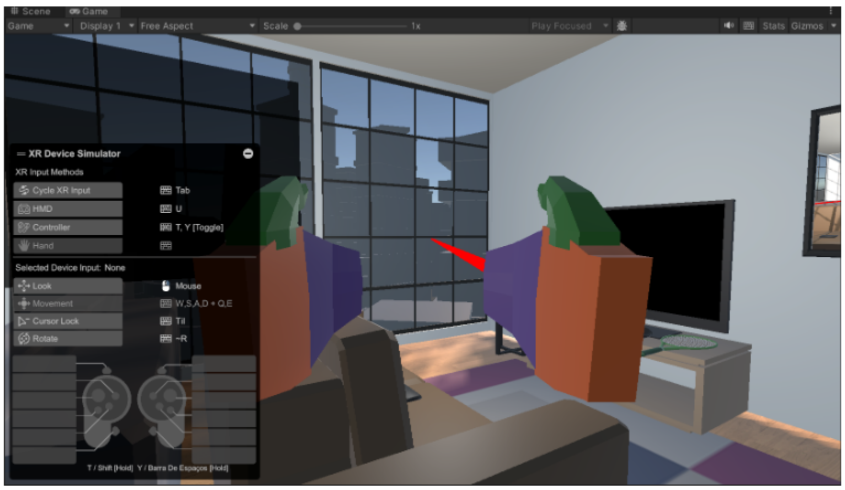

Figure 5: VR Tutorial's Beginner Room Using an XR Simulator

3.4 Creating Puzzles

After finishing the VR tutorials in Unity, I started a tutorial on physics puzzles. 6 This tutorial was recommended to be completed before starting the escape room tutorial, so I went ahead and completed it.7

The objective is to create a puzzle where the ball rolls down a ramp or ramps to a designated spot. However, the goal for this tutorial is not only to learn how to create the puzzles, but also learning how to change camera positions. The tutorial walks you through creating your own puzzle, increasing the difficulty, changing the lights, moving the camera, etc. Basically, it was a mechanism to get you more comfortable with the Unity editor and how the games are built.

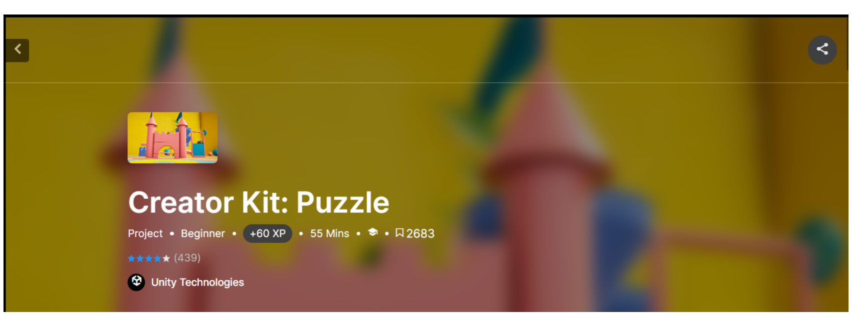

Figure 6: Unity Learn's Puzzle Tutorial

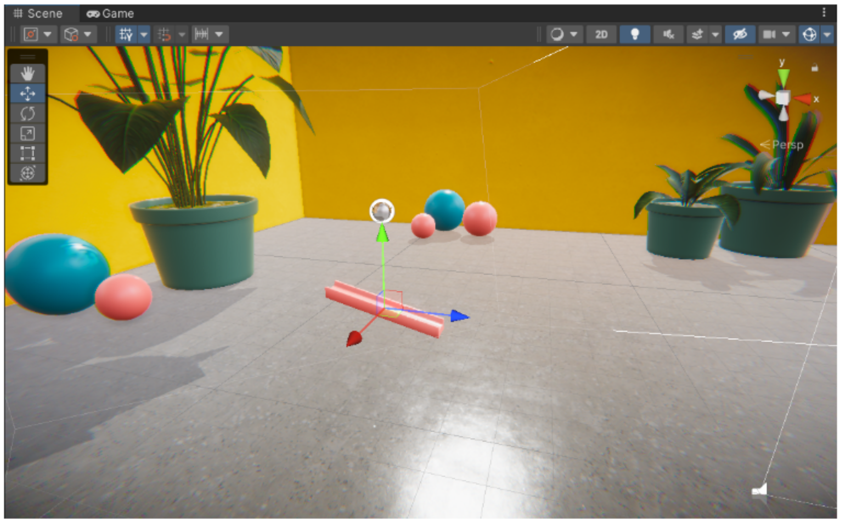

Figure 7: Basic Ramp Physics Puzzle

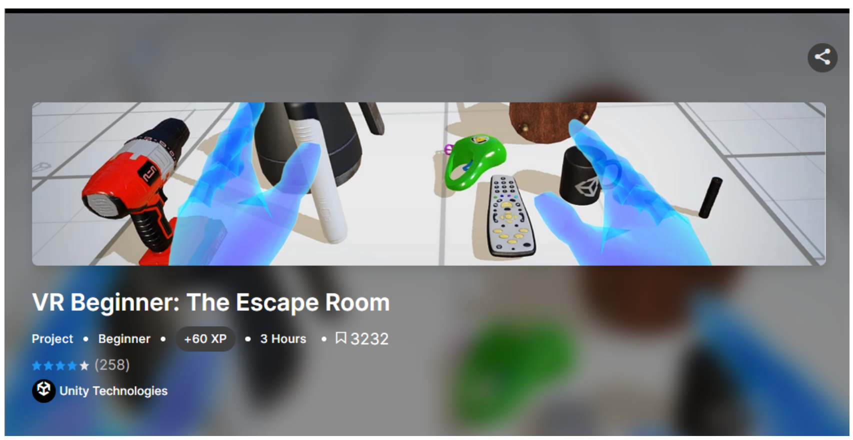

Figure 8: Unity Learn's Escape Room Tutorial

3.5 Escape Room Tutorials and Building My Own

After completing the VR tutorials and getting the headset to work, I started working on the escape room beginning tutorials on Unity Learn. 8 This tutorial pathway walks you through the basics of creating an escape room, interactions in VR, and the exploration of a prebuilt room.

Before you get to the pre-built room, however, Unity walks you back through how to create interactable and grabbable objects in your VE. Using various packages you're able to make it so that you can interact with the puzzles in the room and solve them. You're also walked through adding sound effects, teleportation points, captions, etc. Basically, it gives you the basics on how to create your own escape room. 9

After you've completed the prototype room, you move onto the pre-built room. It's important to note that all of the assets used to create the rooms in this tutorial are included in the asset pack that goes along with these tutorials. [insert citation] You're told to start up your headset and start playing around in the room, and if you get stuck, the tutorial has steps and hints on how to solve your way through it. 10

After just one go in the escape room is when I started to run into problems, again.

Figure 9: Prototype Room in Unity

Figure 10: Pre-Built Escape Room in Unity

3.6 Running Into Problems and Initial Tests

As I finished up the intial tests of the pre-built escape room, I began the process of creating my own from scratch, This is what I consider to be stopping point. At some point during the development process, my headset had stopped running my Unity projects. I had tried everything I could find and had attempted troubleshooting everything I could see. It was eventually revealed to be a hardware problem, but during the unknown period of time, I had tried a multitude of different things. At one point, after trying to just build the project rather than going through the Oculus app, I managed to make sounds play, but after that one glimmer of hope, the headset went back to not working. This would lead to the transition of my project from a development project to a literature review.

3.7 Transitioning to a Literature Review and Final Thoughts on the Development Process

After not being able to continue with development, and working with limited time, I transitioned my project to a literature review. The focus of the literature review is on the intersection of VR and education, along with other applications of immersive education in the forms of escape rooms.

The development process definitely taught me a lot. I also am glad that I chose Unity because I was able to get at least this far. However, during the troubleshooting portions, Unity was little to no help and I had to depend on other sources to get my headset to run even my initial VR projects. So, even though Unity is very User-Friendly and beginner cognizant, they don't provide as much support if something goes wrong. That being said, I don't think I would choose Unreal over Unity for this specific project. This is mainly because I have never developed anything for VR, and Unity provided the materials I needed as a beginner.

4 Conclusions

This main goal of this project is to create an educational escape room for kids around the late elementary to middle school age. However, after setback and hardware issues, the project has transitioned to a literature review on the intersection between Virtual Reality and Education.

Through the development process I was able to learn about how Unity works and have come to the conclusion that while it is definitely beginner friendly, it is not hardware helpful. When I was going through the troubleshooting process, Unity provided little to no help and I had to rely on other sources to attempt to get the headset working.

After the literature review, I was able to draw a few conclusions.

1. VR has observed benefits in the classroom
2. The immersive nature of VR provides a multitude of ways students can engage with material in the classroom
3. There is a worry about accessibility, both with funding and special needs students
4. Escape Rooms have become increasingly popular in education, but there is still a lack of information on the benefits of using escape rooms in a classroom
5. VR, at least at this point, should be used as a supplemental tool, not the main way of teaching course material
6. Students who have used VR have usually expressed positive reactions from their experiences

Overall, the conclusion that I have come to is that, VR, while is still being innovated, is definitely making its way into the education system and so are escape rooms. While there is still a somewhat lack of research on these topics and their intersection, I think in the future, we can expect Virtual Reality to become part of our education system and practices.

5 Future Work

The overall goal of this project in the future would be to continue the creation of the escape room. The creation process of the escape room was great and I would love to try it on some students. The overall goal of the project as a whole, would be to evaluate the effectiveness of the escape room on kids from around late elementary school to middle school students. After that, I would analyze the data and see if there really is any benefits to Virtual Reality in education, and futhermore, if Virtual Escape Rooms are beneficial or not.

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